

Classwork (23rd August): Joint Probability Distribution

1. In a cloud computing platform, two servers operate simultaneously to handle user requests. During a fixed one-hour monitoring period, let X denote the number of request failures in Server A, and let Y denote the number of request failures in Server B. The joint probability mass function of X and Y is given in the table below:

$Y \setminus X$	0	1
0	0.30	0.20
1	0.10	0.40

- (a) Determine the marginal probability distributions of X and Y .
- (b) Compute the conditional probability that Server A has a failure given that Server B has no failures.
- (c) Determine whether the failures of the two servers are independent.
2. In a wireless communication system, signal quality is affected by both signal attenuation and noise interference. Let X represent the signal attenuation level and Y represent the noise intensity, where both variables take values between 0 and 1. Their joint probability density function is defined as

$$f(x, y) = 1; 0 < x < 1, 0 < y < 1.$$

- (a) From this model, find the marginal probability density functions of X and Y .
- (b) Compute the conditional probability that noise intensity is less than 0.5 given that the signal attenuation level is 0.8.
3. In an economic survey, researchers study the relationship between individual income and consumption level. Let X denote normalized income (scaled between 0 and 1) and let Y denote normalized consumption expenditure (also between 0 and 1). The joint probability density function of X and Y is given by

$$f(x, y) = x + y; 0 < x < 1, 0 < y < 1.$$

- (a) Using this model, first determine the normalizing constant to make it a valid joint density function.
- (b) Find the marginal distributions of income and consumption. Verify if $E(XY) = E(X)E(Y)$.
- (c) Determine whether income and consumption appear to be independent.

4. In a particle physics experiment, detectors measure two quantities during each event. Let X represent the time (in microseconds) until a particle is detected and let Y represent the normalized energy level of the detected particle. The joint density function is given by

$$f(x, y) = e^{-(x+y)}; x > 0, y > 0.$$

- Derive the marginal density functions of X and Y . Find $E(X)$ and $E(Y)$.
 - Compute the probability that the energy level exceeds 1 given that the detection time is at least 1 microsecond.
 - Comment on whether the detection time and energy level appear to be independent.
5. In a manufacturing process, two quality indicators are observed for each produced item. Let X denote the number of minor defects per item (taking values 0, 1, or 2) and let Y denote the severity index of defects, which is a continuous variable between 0 and 1. Their joint distribution is defined by

$$f(x, y) = c(x + 1)y; x = 0, 1, 2 \text{ and } 0 < y < 1.$$

- Determine the value of c that makes this a valid joint probability model.
(Hint: X is a discrete random variable and Y is a continuous random variable)
 - Find the marginal distributions of X and Y .
 - Compute the probability that the severity index exceeds 0.5 given that the item has two defects.
6. In a wireless communication system, two continuous random variables are observed during signal transmission. Let X denote the signal attenuation time (in seconds) before the signal drops below a threshold and let Y denote the noise interference level (normalized between 0 and 1) during transmission. Due to system design constraints, X cannot exceed 5 seconds and Y is bounded between 0 and 1. The joint cumulative distribution function (CDF) is given by:

$$F(x, y) = \begin{cases} \frac{x^2 y}{25}; & 0 \leq x \leq 5, 0 \leq y \leq 1 \\ 1; & x > 5, y > 1 \end{cases}$$

- Derive the joint probability density function $f(x, y)$ from the given joint CDF.
- Find the marginal density function of X , $f_X(x)$ using the PDF derived in (a). Find the mean signal attenuation time.
- Given that the signal attenuation time is at least 3 seconds, find the probability that the noise interference level is at least 0.5.

$$\text{Hint: } f_{Y|X}(y \geq 0.5 | x \geq 3) = \frac{f_{X,Y}(x \geq 3, y \geq 0.5)}{f_X(x \geq 3)}$$